ANSWER BOOK

Engineering Project Management & Finance

UESTC3031 & UESTCHN3012

(2024 – 2025 Semester 2)

Enhancing Engineering Management Learning: Soft Skills Development: Answer File

Date Issued to Students:

Chengdu Campus: *14th March 2025 (Friday)*
Hainan Campus: *27th March 2025 (Thursday)*

Deadline / Submission Date:

Chengdu Campus: *28th March 2025 (Friday, 11:59 PM CST)*
Hainan Campus: *10th April 2025 (Thursday, 11:59 PM CST)*

Submission: via Moodle

Submission Instructions:

- You must submit in **ONE PDF document** (not a zipped file, not two times, not in a Word document) as the report of this assignment titled **"Enhancing Engineering Management Learning: Soft Skills Development"**, (only 1 submission in a system for each student is allowable).
- You need to complete the assignment report in this provided **Answering Booklet** file maintaining the relevant format.
- You must prepare this assessment **individually (not in a group/team)**. In the file name, indicate your name and ID in the following format **StudentName_GUID_UESTCID.pdf.** Inside the document, you need to provide the **student's name, GUID, and UESTCID** so that we know who submitted the report.
- You must submit a **plagiarism declaration,** and your submission will go through plagiarism check.
- **No late submissions will be allowed in Moodle** so if you don't submit you will get a zero for this complete assessment.
- This is a **SUMMATIVE assessment** (i.e., marked assessment). The marking criteria are provided.

Student

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Remember to Complete and Sign the Declaration Form on Page 2.

Student: Declaration of Originality Form

This form **must** be completed and signed and submitted with all assignments.
Please complete the information below (using BLOCK CAPITALS).

Name:Yichi Chen.....
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Course Name: Engineering Project Management & Finance.....
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Read and understood the guidance on plagiarism in the Student Handbook, including the University of Glasgow Statement on Plagiarism	√
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Fully referenced (including page numbers) and used inverted commas for **all text quoted** from books, journals, web etc. (Please check with the Department which referencing style is to be used)	√
Provided the sources for all tables, figures, data etc. that are not my own work	√
Not made use of the work of any other student(s) past or present without acknowledgement. This includes any of my own work, that has been previously, or concurrently, submitted for assessment, either at this or any other educational institution, including school (see overleaf at 31.2)	√
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Signed Yichi Chen.....

Enhancing Engineering Management Learning: Soft Skills Development
[100 marks]

Notes:

1- Each solution corresponding to each question must not exceed 250 words.
2- The total mark for this report is based on 100 and will be scaled regarding the second assignment**.**

1. Question 1 [10/100]

👉 **Scenario:** Your company is designing a new smart home device. You have two material options for the outer casing:

- **Option A:** Recyclable biodegradable plastic, but 30% more expensive.
- **Option B:** Standard ABS plastic, lower cost but not environmentally friendly.

As a DFM engineer, how do you approach this decision? What factors do you consider?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Problem Solving, Decision-Making, Clarity of Response (Presentation Skills)

Answer:

choosing the material for the new smart home device is a difficult task. A lot of factors should be taken into consideration. Here are my working processes and factors I consider:

Cost and Future Considerations

- Direct Costs: The biodegradable plastic (Option A) is 30% more expensive than ABS, which is a significant increase. But we can alleviate its influence by charging more for the product or improving the value of our brand.
- Future Risks: Although ABS maybe safer, using it could become costly later on. Predictably, environmental regulations tighten, sticking with ABS could result in higher costs or even fines.

Manufacturability and Supply Chain

- Production Compatibility: It's important to assess whether biodegradable plastic is suitable for our current manufacturing processes., otherwise we could pay great price to make the necessary adjustments.
- Supply Reliability: It is also essential to ensure that the eco-friendly material is reliably available, so that production delays can be avoided.

Environmental Impact and Market Trends

- Sustainability: Option A aligns with circular economy principles and supports our commitment to corporate responsibility.
- Consumer Demand: As the demand for eco-friendly products grows, Option A could help us take advantage of this trend and potentially command a higher price, attracting more customers.

To sum up, while Option B might save money initially, Option A better matches our long-term goals and brand positioning.

2. Question 2 [10/100]

👉 **Scenario:** Your team is producing high-frequency power converters for an EV (Electric Vehicle) charging station. A batch fails thermal stress tests, causing overheating at high loads. The Quality team suggests more testing, while the Manufacturing team wants to modify the heat sink design.

- As the lead engineer, how do you decide?
- How do you communicate with both teams?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Problem Solving, Risk Management, Communication & Stakeholder Engagement & Leadership, Clarity of Response (Presentation Skills)

As the lead engineer, I would made the following decisions to settle this contradiction:

First, to pinpoint root causes (materials, processes, or design)，I would clearly define and analyze the overheating problem using Quality Control tools such as a Cause-and-Effect (Fishbone) diagram and Pareto analysis.

Second, I plan to evaluate the proposed options. In one hand, the Quality team's suggestion of additional testing provides deeper defect insights but delays immediate solutions. In the other hand, the Manufacturing team's heat sink redesign quickly addresses overheating but involves potential risks and costs. Therefore, I'd use Six Sigma's DMAIC (Define, Measure, Analyze, Improve, Control) methodology to systematically assess these options, balancing technical feasibility, risk, cost-effectiveness, and time-to-market.

Moreover, I would communicate transparently with both teams, ensuring alignment and clarity:

- With the Quality team, emphasize how targeted testing aligns with long-term reliability and customer satisfaction, supported by data visualization like Pareto and Control charts.
- With the Manufacturing team, explain how design adjustments using Robust Design principles (Taguchi methods) and optimized material choices can effectively reduce variability, minimize overheating risks, and lower overall costs.

My approach also includes conducting regular joint meetings and feedback sessions, utilizing clear visual presentations (flowcharts, diagrams), to maintain effective stakeholder communication, facilitate collaboration, and ensure timely issue resolution.

3. Question 3 [10/100]

👉 **Scenario:** Your company is designing a high-efficiency PCB (Printed Circuit Board) for sensors used in connected devices, such as smart home appliances or industrial systems. A supplier informs you of ±10% variability in copper trace thickness.

- How would you apply the Taguchi robust design principles to ensure the PCB performs consistently despite this variability?
- How would you collaborate with the supplier to minimize the impact of this variability on the final product?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Teamwork, Problem Solving, Clarity of Response (Presentation Skills)

To handle the ±10% variability in copper trace thickness, I would start by clearly identifying key factors affecting PCB performance. Taguchi robust design principles provide a structured method to analyze factors such as trace thickness, materials, environmental conditions, and manufacturing processes.

I propose to conduct targeted experiments using Taguchi's Orthogonal Arrays. This approach helps evaluate each factor's impact on PCB reliability. These experiments could find the best parameter settings to reduce sensitivity to variability, ensuring consistent performance.

Using the Taguchi Quadratic Loss Function could also help measure the cost associated with performance variations. This quantification guides effective decision-making about managing acceptable variability levels.

For supplier collaboration, clear and transparent communication about shared quality goals is vital. Joint audits of manufacturing processes can identify and address variability causes. Sharing detailed experimental results helps suppliers understand how specific process improvements enhance product quality.

This collaborative approach is essential for minimizing variability, improving production quality, and achieving reliable PCB performance.

4. Question 4 [10/100]

👉 **Scenario:** Your company produces high-precision components for medical devices. The current defect rate is higher than expected, and the process seems to have variability. You are tasked with improving the process to ensure fewer defects and consistent quality.

- How would you assess the performance of the process to identify areas for improvement?
- What steps would you take to reduce defects and increase the process's ability to consistently meet quality standards?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Teamwork, Problem Solving, Presentation Skills (Clarity of Response)

I would start assessing process performance by applying Quality Control tools like Control Charts. These charts help detect variability, trends, or shifts causing defects. Additionally, Pareto analysis clearly highlights the most frequent defect causes, guiding the team to prioritize critical issues.

Below are several specific actions that can effectively reduce defects and improve quality consistency:

1. Apply Six Sigma DMAIC Methodology:
    - Define the quality problem and set clear, measurable improvement goals.
    - Measure current performance by tracking defect rates and calculating process capability.
    - Analyze the process using Cause-and-Effect diagrams to pinpoint root causes of defects.
    - Implement targeted improvements that address these root causes directly.
    - Continuously monitor the improved process to ensure sustained results, regularly updating documentation.
2. Introduce Robust Manufacturing Design:
    - Use Taguchi Robust Design experiments to determine optimal parameters that minimize sensitivity to variability.
3. Establish a Continuous Improvement Culture:
    - Engage employees through regular training and encourage active participation in quality assurance initiatives.

In recap, I intend to ensure clear communication, supported by visual tools and analytical data, reinforcing team alignment and consistently meeting quality standards.

5. Question 5 [10/100]

👉 **Scenario:** Your company is manufacturing resistors, and you are noticing some inconsistencies in the output. You have been tasked with implementing Statistical Process Control (SPC) to monitor and control the production process.

- How would you use control charts to identify potential issues in the production process?
- What actions would you take if you noticed the process is out of control based on the control charts?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Teamwork, Problem Solving, Presentation Skills (Clarity of Response)

Firstly, I would utilize Control Charts from Statistical Process Control (SPC) to monitor the resistor production process over time. Plotting production data on these charts lets me visually check process stability. I can also look for variability or trends in the data. Additionally, I can identify any data points outside the Upper Control Limit (UCL) or Lower Control Limit (LCL). These could signal potential issues.

If the control charts indicate the process is out of control, I would take the following actions:

1. Pause the Production Line Temporarily:
    - Immediately stop production to prevent further defects from occurring.
2. Use a Cause-and-effect (Fishbone) Diagram:
    - Clearly identify and analyze potential root causes, such as materials, machines, human factors, and environmental conditions.
3. Conduct a Pareto Analysis:
    - Prioritize the most critical factors causing defects, enabling focused and effective problem-solving.
4. Implement Six Sigma's DMAIC Methodology:
    - Define the specific quality issues and establish clear improvement objectives.
    - Measure the current defect rate and process capability.
    - Analyze data collected to confirm root causes.
    - Improve the process by addressing identified root causes directly.
    - Control improvements through ongoing monitoring and updating of process procedures.
5. Maintain Regular Monitoring and Communication:
    - Regularly communicate with the production team, reviewing the effectiveness of implemented actions, and continuously monitor the process to ensure sustained control.

6. Question 6 [10/100]

👉 **Scenario:** Your team is working on a project to develop a smart metering system. A stakeholder requests a last-minute addition of AI-based energy prediction features, requiring an extra two months of work.

- How do you handle this request?
- How do you communicate the impact?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

To address the stakeholder's request for adding AI-based energy prediction features requiring two extra months, I plan to :

- Evaluate Impact: Conduct a detailed analysis using Project Scope Management to clearly define scope changes. This involves assessing impacts on schedules, costs, and resources, and systematically documenting the results.
- Risk Management: Leverage Project Risk Management principles to identify potential risks resulting from the extended timeline and additional features. Possible risks include resource limitations, technical uncertainties, and market impacts.
- Stakeholder Communication: Arrange a prompt meeting to present the findings transparently. Clear visuals and data-supported scenarios will illustrate how scope changes impact project timelines, costs, and deliverables.
- Negotiation and Decision Making: Engage stakeholders in structured discussions to evaluate the benefits versus the additional costs and potential delays. Data-driven decision-making techniques and visual aids will support informed discussions.
- Planning and Implementation: Upon stakeholder agreement, update the project plan clearly to ensure alignment. Clearly communicate the revised schedule and resource allocations to all stakeholders.

This Methodical approach guarantees clear communication, thorough evaluation of implications, and informed decisions aligned with stakeholder expectations and overall project objectives.

7. Question 7 [10/100]

👉 **Scenario:** A key embedded systems engineer on your drone navigation project resigns unexpectedly. Your deadline is two months away, and hiring a replacement takes six weeks.

- How do you reallocate resources?
- How do you ensure project deadlines are met?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

To effectively handle the unexpected resignation of the key embedded systems engineer and ensure project deadlines are met, I would take the following steps:

1. Resource Assessment and Reallocation:
    - Quickly identify tasks and responsibilities previously handled by the departing engineer.
    - Assess team skills and availability to temporarily redistribute critical tasks, ensuring minimal disruption.
    - Prioritize tasks based on project milestones and allocate resources accordingly to keep essential activities on track.
2. Risk Management and Mitigation:
    - Conduct immediate risk analysis to identify potential impacts on deadlines and project quality.
    - Develop contingency plans, such as outsourcing specialized tasks or utilizing consultants to cover skill gaps.
3. Communication and Stakeholder Management:
    - Clearly inform stakeholders about the situation, outlining potential impacts and proposed solutions.
    - Regularly update stakeholders on progress, resource reallocation decisions, and timelines.
4. Accelerated Recruitment and Onboarding:
    - Initiate a fast-tracked hiring process immediately, leveraging professional networks and recruiting resources.
    - Prepare for expedited onboarding to rapidly integrate the new engineer into the project team, minimizing disruption.

It is believed that, through proactive resource management, clear communication, thorough risk assessment, and accelerated hiring strategies, project deadlines and objectives can remain achievable despite unforeseen personnel changes.

8. Question 8 [10/100]

👉 **Scenario:** Your company is developing a 5G signal amplifier. The project is over budget by 15%, and management asks for cost reductions without reducing performance.

- How do you identify cost-cutting areas?
- How do you justify your decisions to management?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Decision-Making, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

To identify cost-cutting areas without compromising performance I would conduct a comprehensive review of spending and processes:

1. Perform Detailed Cost Analysis:
    - Analyze current expenditures thoroughly to identify non-critical costs, waste, and inefficiencies.
    - Utilize Pareto analysis to pinpoint high-impact cost-saving opportunities, focusing on areas like procurement, manufacturing processes, and overheads.
2. Implement Value Engineering Principles:
    - Evaluate all product features and materials systematically to determine their cost versus functional benefit.
    - Identify alternatives offering similar performance at lower costs, such as sourcing from alternative suppliers or adopting leaner manufacturing methods.
3. Engage Cross-functional Teams:
    - Collaborate with engineers, procurement specialists, and manufacturing teams to brainstorm viable cost-reduction solutions.

Justifying these decisions to management involves:

- Data-driven Analysis: Clearly present cost-benefit analyses demonstrating minimal impact on performance and significant cost savings.
- Risk Management: Provide comprehensive risk assessments, clearly outlining any potential impacts and mitigation strategies.
- Transparent Communication: Employ visual presentations and straightforward explanations to build consensus and demonstrate thoughtful, performance-focused cost reduction.

This structured approach ensures that cost reductions are strategic, data-supported, and meet project and organizational objectives.

9. Question 9 [10/100]

👉 **Scenario:** Your team is working on a high-priority software development project with tight deadlines. The project schedule is falling behind due to resource constraints, particularly a shortage of skilled developers. Management asks you to expedite the project delivery without compromising quality.

- How do you assess the impact of resource constraints on the project schedule?
- What steps would you take to manage both the schedule and resources efficiently?

💡 ***Soft Skills Assessed:*** Analytical Thinking & Project Management, Critical Thinking, Communication & Stakeholder Negotiation, Risk Management, Presentation Skills (Clarity of Response)

To assess the impact of resource constraints on the project schedule, I would:

1. Perform Schedule Analysis:
    - Utilize project management software and techniques like Critical Path Analysis to identify tasks most affected by resource shortages.
    - Evaluate how shortages impact project milestones, deliverables, and overall timelines.
2. Conduct Resource Analysis:
    - Clearly identify specific skill gaps and quantify the shortage's impact on productivity and timelines.
    - Analyze resource allocation and workload distribution to pinpoint inefficiencies.

To efficiently manage both schedule and resources, I would:

1. Prioritize and Reallocate Resources:
    - Prioritize tasks critical to project success and strategically redistribute existing skilled resources to these tasks.
2. Consider temporary hiring:
    - Explore temporary outsourcing or hiring contract developers to fill immediate skill gaps without compromising project quality.
3. Enhance Productivity:
    - Implement productivity-enhancing practices such as agile methodologies or time-boxing, improving efficiency without reducing quality.
4. Maintain Transparent Communication:
    - Clearly communicate challenges, proposed actions, and progress regularly to management and stakeholders to maintain transparency and alignment.

Adopting these focused strategies supports effective resource management, precise schedule control, and ensures the project meets its deadlines while upholding high quality standards.

10. Question 10 [10/100]

👉 **Scenario:** During your project, two key stakeholders—one from the client side and one from the development team, have a significant disagreement. The client insists on additional features being included, which will extend the project timeline and increase costs, while the development team is concerned about the feasibility and potential impact on existing functionality. As the project leader, you must resolve this conflict without disrupting the project's progress.

- How do you manage this conflict between the two stakeholders while maintaining a positive team dynamic and client relationship?
- How do you ensure that both parties' concerns are addressed and that the project remains on track?

💡 ***Soft Skills Assessed:*** Leadership & Teamwork, Conflict Management, Communication & Stakeholder Negotiation, Communication & Emotional Intelligence, Presentation Skills

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I believe that managing stakeholder conflict requires a balance of empathy, structure, and transparency. To address this situation, I would take the following steps:

1. **Conflict Management and Relationship Preservation**:
    - Conduct individual meetings with both stakeholders to thoroughly understand their perspectives and concerns.
    - Facilitate a structured, collaborative discussion allowing each stakeholder to voice their concerns in a respectful and productive environment.
    - Focus on interests rather than positions to identify common goals and areas of agreement.
    - Use active listening and empathy to maintain trust and a positive relationship with both parties.
2. **Ensuring Both Parties' Concerns are Addressed and Project Progress**:
    - Clearly outline the impacts of requested changes, including timelines, costs, and technical feasibility, supported by objective data.
    - Collaboratively explore alternative solutions or compromises that meet core requirements without significant negative impact.
    - Clearly document and communicate agreed-upon decisions and action plans to ensure alignment and transparency.
    - Establish clear follow-up meetings to monitor progress, ensuring that agreements are honored, and adjustments can be made promptly if necessary.

I intend to approach this conflict constructively, ensuring both voices are respected while steering the project forward. My goal is to maintain a positive dynamic and deliver a successful outcome for all stakeholders involved.